

Authenticity Metric - Appendix

Atomic Intelligence R&D team

Appendix C: Synthetic Data Privacy

When estimating the quality of synthetic data, researchers usually focus on three or three independent metrics, fidelity, utility, and privacy.

Fidelity focuses on measuring how well the synthetic data reproduce the patterns present in the data. It can be further broken down into real fidelity and generalization. Real fidelity measures whether all of the synthetic data are somehow close to the real data, while generalization measures how much of the real data feature space the synthetic data *covers*. The authors [1] provide clear and interpretable α - *precision* and beta β - *recall* measures that estimate real fidelity and generalization, these, however, have not been part of our fidelity analysis.

Utility measures how well the synthetic data can be used in downstream tasks compared to the real data, and as such it is harder to cover the entire spectrum of possible use cases with a single metric. High-fidelity scores should also imply high utility scores, but so far there are no theoretical guarantees. The most common approach to estimating utility involves calculating the "machine learning efficiency" score on synthetic data. This process typically requires constructing classification or regression tasks on both real and synthetic datasets, training a simple model on the real data, and then evaluating its performance on synthetic data. Ideally, similar performance scores should be achieved across both datasets. This evaluation can be further refined by training multiple models on the real data and verifying that not only are the metrics comparable between real and synthetic data, but also that the relative performance ranking of the models remains consistent across both datasets.

Both high fidelity and utility can be achieved by simply copying real data and adding some small noise to it. This is clearly undesirable, as synthetic data is typically used in scenarios where utilizing real data is prohibited for various reasons. Additionally, synthetic data may contain other vulnerabilities that malicious actors could potentially exploit. To address these concerns, researchers measure the *privacy* of synthetic data. While various definitions and approaches exist for estimating privacy protection, our focus is primarily on data authenticity

Measures such as Membership Identification Attacks (MIA) and Re-Identification risks [2] are often considered the gold standard for estimating data privacy. Here, a discriminative model is trained to distinguish between real data used for training a synthetic data generator and real data that was not used for training the generator. Note that the discriminator is trained on exposed synthetic data and real data not used for training the generator. The accuracy of the discriminator is considered the membership identification risk; the closer to random guessing the accuracy becomes, the more private the data. While this approach is applicable in many situations, we believe it greatly depends on the selected model, and estimating privacy using the current state-of-the-art

methods says very little on possible vulnerabilities the exposed synthetic data may have against future methods.

Instead, we have chosen a model-agnostic approach based on the Nearest-Neighbors (NN) algorithm.

Appendix C1: Authenticity metric

The key property of the authenticity metric should be that it measures whether the synthetic data generation algorithm simply copies the real data with some added noise. Low authenticity scores should correspond to high degrees to *copied* data. The base assumption is that the synthetic data distribution $\mathbb{P}_g$ is decomposed as [1]

$$\mathbb{P}_g = A \cdot \mathbb{P}'_g + (1 - A) \cdot \delta_{g,\epsilon}$$

Here $\mathbb{P}'_g$ is the distribution of the synthetic data that generates authentic examples, and $\delta_{g,\epsilon}$ corresponds to the part of the total synthetic data distribution which *memorizes* training (real) examples and copies them with some small added noise. This corresponds to sampling from a discrete distribution δ_g which corresponds to a copied subset of the real training dataset, convolved with a normal distribution with variance ϵ^2 .

$$\delta_{g,\epsilon} = \delta_g * \mathcal{N}(0, \epsilon^2)$$

Lastly, A is the probability with which the total distribution of synthetic data $\mathbb{P}_g$ generates authentic samples.

The goal is to estimate A in some way to get a measure of authenticity of our data, and we do this by using the distance-to-closest-record metric (DCR).

1. We embed the data (both real and synthetic) into some latent feature space.
Note, that the embedding need not be to some abstract feature space, it can also refer to data scaling and handling categorical features. Furthermore, if embedding functions require any sort of fitting, it is done only on the real data.
2. For each example of the synthetic data $X_{g,j}$, we calculate the distance to its closest neighbor $X_{r,k}$ among the real data $d(X_{g,j}, X_{r,k}) = d_{j,k}$.
3. Next, for the closest real neighbor $X_{r,k}$, we find the distance $d_{k,i}$ between it and its closest real neighbor $X_{r,i}$.
4. Lastly, we define the binary indicator variable $a_j = \mathbf{1}\{d_{k,i} < d_{j,k}\}$.

The variable a_j indicates whether a synthetic data point is farther from its closest real data neighbor than at least one real data point is from that closest real neighbor. By averaging these a_j values, we obtain an estimate of A , the overall authenticity measure.

The authors [1] argue that mean $\langle a \rangle$ is a good proxy for estimating A in the limit where the size of the real (training) dataset becomes infinite. They do not, however, give any numerical or theoretical bounds on the size of the real data set for the mean $\langle a \rangle$ to be a good estimate of A . Because of that, instead of reporting only the average authenticity, we also report on the closest neighbor distance ratio (CNDR) r_j defined as

$$r_j = \frac{d_{j,k}}{d_{k,i}}$$

Note that authentic examples will have $r_j \geq 1$, while inauthentic examples will have $r_j < 1$. We also report on the distribution of CNDR of the synthetic dataset. This allows us to even soften the definition of authentic examples as ones with a r_j above some threshold. We decided against that in order to reduce the possible noise in our analysis.

Appendix C2: Authenticity results

We estimate the DCR and CNDR of the synthetic data, mainly for the peptide features and clinical features generated with the Gaussian Copula model. Numerical features are scaled to unit interquantile range (IQR) and zero median while categorical features are one-hot encoded. In this space of scaled numerical features and encoded categorical features the nearest neighbor algorithm was performed.

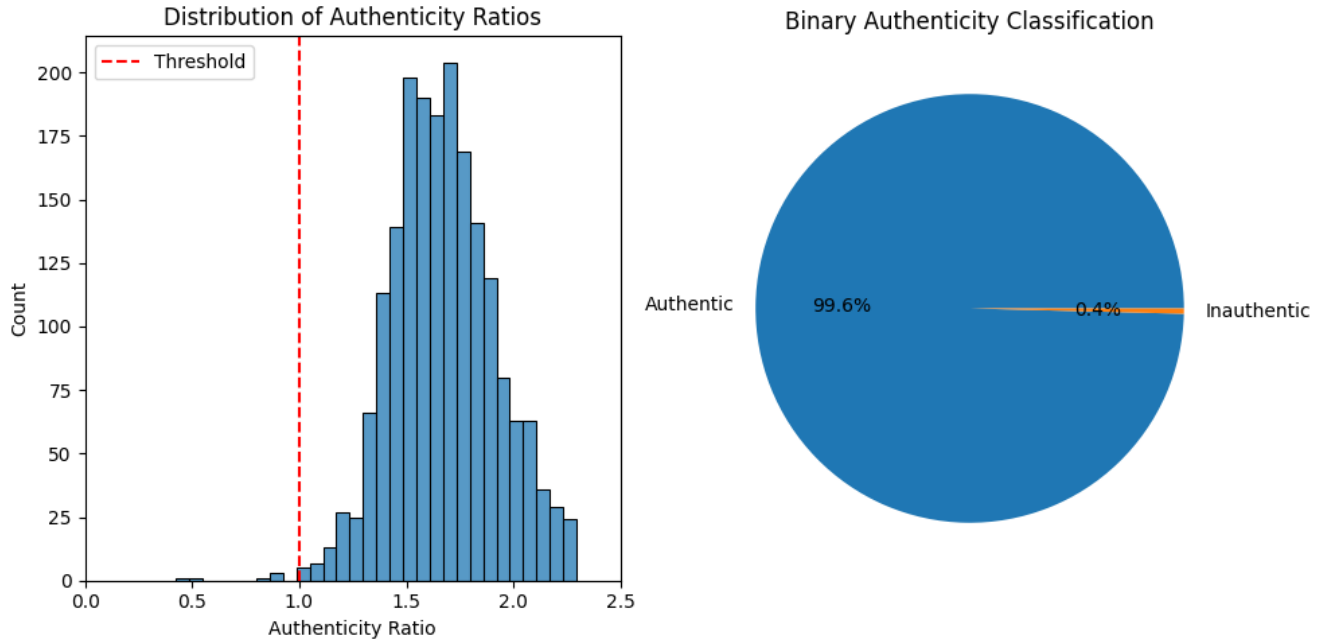


Figure 1: Authenticity results for peptides and clinical features generated with the Gaussian Copula model. The image on the left shows the closes neighbor distance ratio for every synthetic data example. The image on the right shows the binary authenticity of the synthetic data, this corresponds to $A = 0.996$.

The figure 1 summarizes the results of our analysis. Only 0.4% of the generated examples were closer to a real record than any other real record, which we deem inauthentic. This in of itself is a satisfactory result; however, we decided to explore the CNDR as well. The left image in 1 we can see the distribution of these CNDR or *authenticity ratios* (r_j). We find that even examples which fall below the authenticity threshold (in this case 1.0) are not significantly closer to a real record than the next closest real record. This can be interpreted as the risk of our algorithm copying real data being small.

Appendix C3: Interpreting privacy with equivalent noise

The mean authenticity metric provides some intuitive insight as to how much the synthetic data generation mechanism copies real data; however, it is not immediately obvious how much privacy our generation mechanism guarantees.

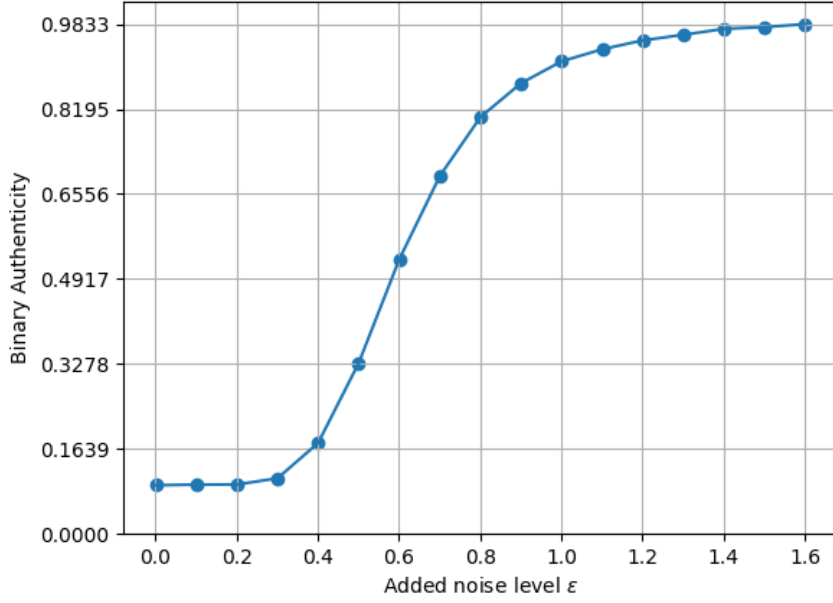


Figure 2: Authenticity levels for varying levels of add noise to the data. We perturb the features of the real data with some Gaussian noise proportional to ϵ and estimate the binary authenticity corresponding to that noise level.

We offer an interpretation that we call the *equivalent noise*. The premise is to assume the worst-case scenario and assume that our synthetic data-generating mechanism does indeed memorize real data examples and adds some noise to it to generate synthetic data. The question is now how much noise should be added to obtain the reported levels of authenticity? To answer this, we generate synthetic data by randomly choosing real data examples and adding random noise sampled from $\mathcal{N}(0, \epsilon^2 \sigma_i^2)$ to each feature, where σ_i is the standard deviation of the i -th feature. Once this perturbed data is generated we calculate the mean binary authenticity on it. This process is repeated for varying levels of noise ϵ .

We are left with a relation that connects the mean binary authenticity to the noise level added to the data. From that relation we can get some sort of bound on the data privacy, i.e. our generated data is at least as private as adding noise ϵ^* that corresponds to the authenticity A^* of our synthetic data w.r.t the DCR metric.

Figure 2 shows how the binary authenticity (or DCR) metric changes with the added noise level ϵ to the real dataset. So, the worst-case scenario for privacy would be for our algorithm to copy data and add noise to it. From the graph in figure 2 we can see that in order to achieve the $A = 0.996$ we would need noise levels well above $\epsilon = 1.5$. Such high levels of noise would correspond to a strong *blurring* effect. The conclusion is that even the worst case of non-private algorithms would require high levels of added noise to achieve our reported levels of authenticity.

References

1. Alaa, A. M., van Breugel, B., Saveliev, E. & van der Schaar, M. *How Faithful is your Synthetic Data? Sample-level Metrics for Evaluating and Auditing Generative Models* 2022. arXiv: [2102.08921](https://arxiv.org/abs/2102.08921) [cs.LG]. <https://arxiv.org/abs/2102.08921>.
2. Carey, C. *et al. Measuring Re-identification Risk* 2023. arXiv: [2304.07210](https://arxiv.org/abs/2304.07210) [cs.CR]. <https://arxiv.org/abs/2304.07210>.